

PHASE 4

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In this phase, we implemented our idea of football players with basketball players so that we are able to provide similar system where user can do as they wish with both collections adding or updating data etc. Our collection has their attributes such as name, surname ,position weight, points. Points are where users give their points to their player that they wanna add.

1. Create Collection

Here we created collections basketball and football players via request from user with dummy data we had. Showed on Python that is succesfully created and showed database afterwards.

```
● $ python main.py
  Select an operation:
  1 - Create a collection
  2 - Read all data in a collection
  3 - Read some part of the data while filtering
  4 - Insert data
  5 - Delete data
  6 - Update data
  Enter your choice: 1
  Enter collection name: football_players
  Collection created successfully.
● $ python main.py
  Select an operation:
  1 - Create a collection
  2 - Read all data in a collection
  3 - Read some part of the data while filtering
  4 - Insert data
  5 - Delete data
  6 - Update data
  Enter your choice: 1
  Enter collection name: basketball_players
  Collection created successfully.
```

Collection creation

localhost:27017

My Queries

Performance

Databases

Search

admin

config

las

local

test

My Queries

Databases

Create database

Refresh

admin

Storage size:

20.48 kB

Collections:

0

Indexes:

1

config

Storage size:

36.86 kB

Collections:

0

Indexes:

2

las

Storage size:

28.67 kB

Collections:

3

Indexes:

3

local

Storage size:

45.06 kB

Collections:

1

Indexes:

1

test

Storage size:

73.73 kB

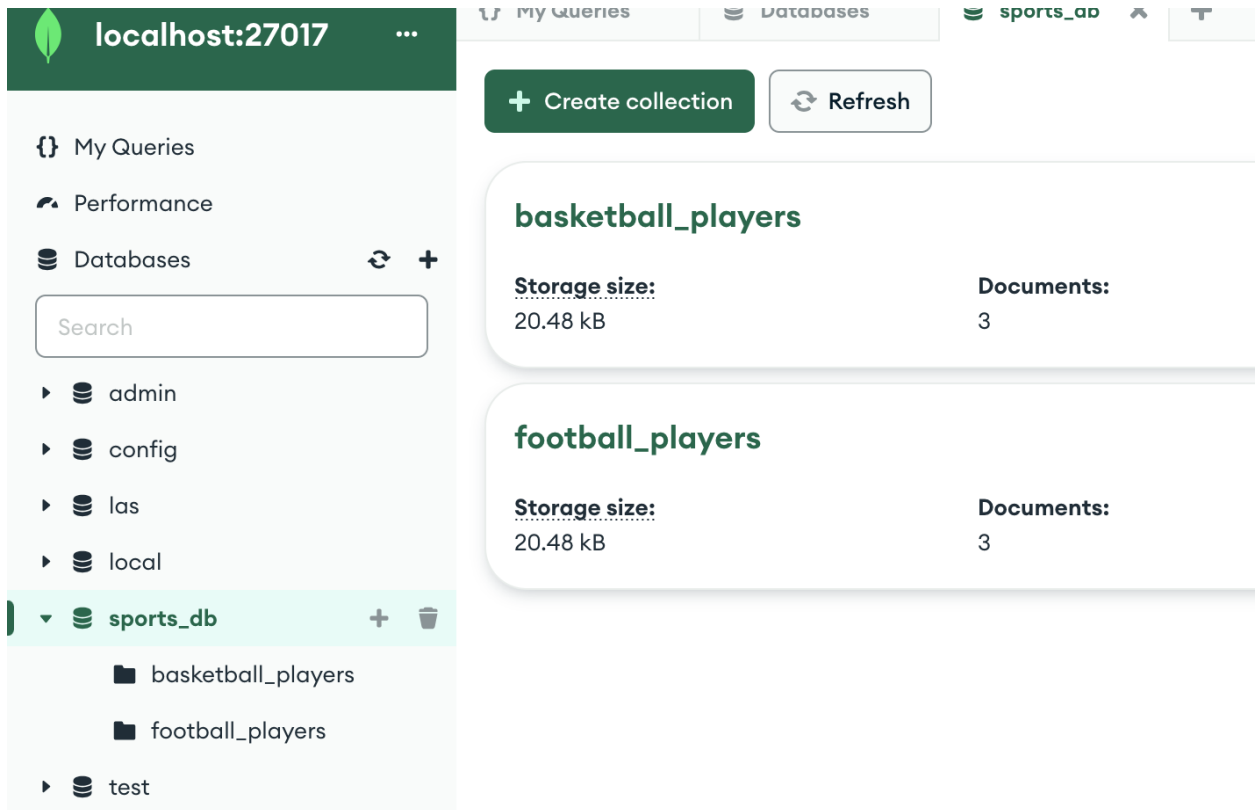
Collections:

2

Indexes:

2

Db before creation



Db after creation

2. Read all data in collection

Here we read the data we head in the Python for both basketball and football players.

```

$ python main.py
Select an operation:
1 - Create a collection
2 - Read all data in a collection
3 - Read some part of the data while filtering
4 - Insert data
5 - Delete data
6 - Update data
Enter your choice: 2
Choose the collection:
1 - Football players
2 - Basketball players
Enter your choice: 1
{'_id': ObjectId('66571334a6bebfb31ac7cf9'), 'name': 'Lionel', 'surname': 'Messi', 'height': 1.7, 'weight': 72, 'position': 'Forward', 'average_point': 9.5, 'points': [10, 9]}
{'_id': ObjectId('66571334a6bebfb31ac7cfa'), 'name': 'Cristiano', 'surname': 'Ronaldo', 'height': 1.87, 'weight': 83, 'position': 'Forward', 'average_point': 9.3, 'points': [9, 10]}
{'_id': ObjectId('66571334a6bebfb31ac7cfb'), 'name': 'Neymar', 'surname': 'Jr', 'height': 1.75, 'weight': 68, 'position': 'Forward', 'average_point': 8.7, 'points': [8, 9.4]}
$ python main.py
Select an operation:
1 - Create a collection
2 - Read all data in a collection
3 - Read some part of the data while filtering
4 - Insert data
5 - Delete data
6 - Update data
Enter your choice: 2
Choose the collection:
1 - Football players
2 - Basketball players
Enter your choice: 2
{'_id': ObjectId('665713692983ef418e786282'), 'name': 'LeBron', 'surname': 'James', 'height': 2.06, 'weight': 113, 'position': 'Small Forward', 'average_point': 9.8, 'points': [10, 9.6]}
{'_id': ObjectId('665713692983ef418e786283'), 'name': 'Stephen', 'surname': 'Curry', 'height': 1.91, 'weight': 84, 'position': 'Point Guard', 'average_point': 9.2, 'points': [9, 9.4]}
{'_id': ObjectId('665713692983ef418e786284'), 'name': 'Kevin', 'surname': 'Durant', 'height': 2.08, 'weight': 109, 'position': 'Small Forward', 'average_point': 9.4, 'points': [9.5, 9.3]}
$

```

Data read of basketball and football players

3. Reading data with filter

This part we showed filtered reading where we were able to filter for position for basketball players and name for football players name showed the filtered data.

```

$ python main.py
Select an operation:
1 - Create a collection
2 - Read all data in a collection
3 - Read some part of the data while filtering
4 - Insert data
5 - Delete data
6 - Update data
Enter your choice: 3
Choose the collection:
1 - Football players
2 - Basketball players
Enter your choice: 1
Enter filter criteria (e.g., {'position': 'Forward'}): {'name': 'Cristiano'}
{'_id': ObjectId('66571334a6bebfb31ac7cfa'), 'name': 'Cristiano', 'surname': 'Ronaldo', 'height': 1.87, 'weight': 83, 'position': 'Forward', 'average_point': 9.3, 'points': [9, 10]}
$ python main.py
Select an operation:
1 - Create a collection
2 - Read all data in a collection
3 - Read some part of the data while filtering
4 - Insert data
5 - Delete data
6 - Update data
Enter your choice: 3
Choose the collection:
1 - Football players
2 - Basketball players
Enter your choice: 2
Enter filter criteria (e.g., {'position': 'Forward'}): {'position': 'Small Forward'}
{'_id': ObjectId('665713692983ef418e786282'), 'name': 'LeBron', 'surname': 'James', 'height': 2.06, 'weight': 113, 'position': 'Small Forward', 'average_point': 9.8, 'points': [10, 9.6]}
{'_id': ObjectId('665713692983ef418e786284'), 'name': 'Kevin', 'surname': 'Durant', 'height': 2.08, 'weight': 109, 'position': 'Small Forward', 'average_point': 9.4, 'points': [9.5, 9.3]}
$

```

4. Inserting Data

For this part we inserted brand new data to our collections and showed it on db as well as Python application. We uploaded players with users desired points for the player that they are adding.

```

● $ python main.py
Select an operation:
1 - Create a collection
2 - Read all data in a collection
3 - Read some part of the data while filtering
4 - Insert data
5 - Delete data
6 - Update data
Enter your choice: 4
Choose the collection:
1 - Football players
2 - Basketball players
Enter your choice: 1
Enter player's name: Declan
Enter player's surname: Rice
Enter player's height: 1.85
Enter player's weight: 79
Enter player's position: Midfielder
Enter point for the player: 7.8
New player inserted successfully.
⊗ $ 4
zsh: command not found: 4
⊗ $ main
zsh: command not found: main
● $ python main.py
Select an operation:
1 - Create a collection
2 - Read all data in a collection
3 - Read some part of the data while filtering
4 - Insert data
5 - Delete data
6 - Update data
Enter your choice: 4
Choose the collection:
1 - Football players
2 - Basketball players
Enter your choice: 2
Enter player's name: Cedi
Enter player's surname: Osman
Enter player's height: 2.07
Enter player's weight: 102
Enter player's position: Small Forward
Enter point for the player: 5
New player inserted successfully.

```

Inserting

sports_db.football_players

Documents

Aggregations

Schema

Indexes

Validation

Filter 

Type a query: { field: 'value' } or [Generate query](#) ✨

Generate query ✨

[+ ADD DATA](#)

EXPORT DATA

```
_id: ObjectId('66571334a6bebfbe31ac7cf9')
name: "Lionel"
surname: "Messi"
height: 1.7
weight: 72
position: "Forward"
average_point: 9.5
points: Array (2)
```

```
_id: ObjectId('66571334a6bebfbe31ac7cfa')
name: "Cristiano"
surname: "Ronaldo"
height: 1.87
weight: 83
position: "Forward"
average_point: 9.3
points: Array (2)
```



```
_id: ObjectId('66571334a6bebfbe31ac7cfb')  
name: "Neymar"  
surname: "Jr"  
height: 1.75  
weight: 68  
position: "Forward"  
average_point: 8.7  
points: Array (2)
```

Before insert

sports_db.football_players

[Documents](#)[Aggregations](#)[Schema](#)[Indexes](#)[Validation](#)[Filter](#)Type a query: { field: 'value' } or [Generate query](#)[+ ADD DATA](#)[EXPORT DATA](#)

```
_id: ObjectId('66571334a6bebfbe31ac7cf9')
name: "Lionel"
surname: "Messi"
height: 1.7
weight: 72
position: "Forward"
average_point: 9.5
points: Array (2)
```

```
_id: ObjectId('66571334a6bebfbe31ac7cfa')
name: "Cristiano"
surname: "Ronaldo"
height: 1.87
weight: 83
position: "Forward"
average_point: 9.3
points: Array (2)
```

```
_id: ObjectId('66571334a6bebfbe31ac7cfb')
name: "Neymar"
surname: "Jr"
height: 1.75
weight: 68
position: "Forward"
average_point: 8.7
points: Array (2)
```

```
_id: ObjectId('665716cba0b6f2e740febd50')
name: "Declan"
surname: "Rice"
height: 1.85
weight: 79
position: "Midfielder"
points: Array (1)
average_point: 7.8
```

After insert

5.Delete Data

Then we showed for our collections that we can delete data if chosen and show it on the database.

```
● $ python main.py
Select an operation:
1 - Create a collection
2 - Read all data in a collection
3 - Read some part of the data while filtering
4 - Insert data
5 - Delete data
6 - Update data
Enter your choice: 5
Choose the collection:
1 - Football players
2 - Basketball players
Enter your choice: 1
Enter filter criteria for deletion (e.g., {'name': 'Lionel'}): {'name': 'Neymar'}
● $ python main.py
Select an operation:
1 - Create a collection
2 - Read all data in a collection
3 - Read some part of the data while filtering
4 - Insert data
5 - Delete data
6 - Update data
Enter your choice: 5
Choose the collection:
1 - Football players
2 - Basketball players
Enter your choice: 2
Enter filter criteria for deletion (e.g., {'name': 'Lionel'}): {'name': 'Stephen'}
Data deleted successfully.
○ $ █
```

Delete

My Queries

Databases

sports_db

basketball_players

test

sports_db.basketball_players

Documents

Aggregations

Schema

Indexes

Validation

Filter

Type a query: { field: 'value' } or [Generate query](#)

+ ADD DATA

EXPORT DATA

_id: ObjectId('665713692983ef418e706282')

name: "LeBron"

surname: "James"

height: 2.06

weight: 113

position: "Small Forward"

average_point: 9.8

points: Array (2)

_id: ObjectId('665713692983ef418e706284')

name: "Kevin"

surname: "Durant"

height: 2.08

weight: 109

position: "Small Forward"

average_point: 9.4

points: Array (2)

_id: ObjectId('6657171fc5ffc83fce92096c')

name: "Cedi"

surname: "Osman"

height: 2.07

weight: 102

position: "Small Forward"

points: Array (1)

average_point: 5

Curry Deleted from Basketball Players

My Queries

Databases

sports_db

football_players

test

sports_db.football_players

DocumentsAggregationsSchemaIndexesValidation

FilterType a query: { field: 'value' } or [Generate query](#)

ADD DATA

EXPORT DATA

```
_id: ObjectId('66571334a6bebfbe31ac7cf9')
name: "Lionel"
surname: "Messi"
height: 1.7
weight: 72
position: "Forward"
average_point: 9.5
points: Array (2)
```

```
_id: ObjectId('66571334a6bebfbe31ac7cfa')
name: "Cristiano"
surname: "Ronaldo"
height: 1.87
weight: 83
position: "Forward"
average_point: 9.3
points: Array (2)
```

```
_id: ObjectId('665716cba0b6f2e740febd50')
name: "Declan"
surname: "Rice"
height: 1.85
weight: 79
position: "Midfielder"
points: Array (1)
average_point: 7.8
```

Neymar deleted from Football Players

6.Update Data

Then at the end we showed that we can update both collections data. We updated names of Lionel and Lebron to GOAT.

```

● $ python main.py
Select an operation:
1 - Create a collection
2 - Read all data in a collection
3 - Read some part of the data while filtering
4 - Insert data
5 - Delete data
6 - Update data
Enter your choice: 6
Choose the collection:
1 - Football players
2 - Basketball players
Enter your choice: 1
Enter filter criteria for update (e.g., {'name': 'Lionel'}): {'name': 'Lionel'}
Enter update values (e.g., {'position': 'midfielder'}): {'name': 'GOAT'}
● $ python main.py
Select an operation:
1 - Create a collection
2 - Read all data in a collection
3 - Read some part of the data while filtering
4 - Insert data
5 - Delete data
6 - Update data
Enter your choice: 6
Choose the collection:
1 - Football players
2 - Basketball players
Enter your choice: 2
Enter filter criteria for update (e.g., {'name': 'Lionel'}): {'name': 'LeBron'}
Enter update values (e.g., {'position': 'midfielder'}): {'name': 'GOAT'}
Data updated successfully.
$

```

Update

sports_db.basketball_players

Documents

Aggregations

Schema

Indexes

Validation

Filter 

Type a query: { field: 'value' } or [Generate query](#) ✦

+ ADD DATA

EXPORT DATA

```
_id: ObjectId('665713692983ef418e706282')  
name: "GOAT"  
surname: "James"  
height: 2.06  
weight: 113  
position: "Small Forward"  
average_point: 9.8  
► points: Array (2)
```

```
_id: ObjectId('665713692983ef418e706284')  
name: "Kevin"  
surname: "Durant"  
height: 2.08  
weight: 109  
position: "Small Forward"  
average_point: 9.4  
► points: Array (2)
```

Goat James

[Documents](#) [Aggregations](#) [Schema](#) [Indexes](#) [Validation](#)

Type a query: { field: 'value' } or [Generate query](#) ✨

EXPORT DATA ▼

```
_id: ObjectId('66571334a6bebfb3e31ac7cfa')  
name: "Cristiano"  
surname: "Ronaldo"  
height: 1.87  
weight: 83  
position: "Forward"  
average_point: 9.3  
► points: Array (2)
```

Goat Messi